Lab Report: Week 2 - Spring Constant

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Aim

To find the spring constant of the spring provided using two different methods.

Theory

We start with Hooke's Law:

$$F = kx$$

Where F = Force, k = spring constant, and x = displacement.

For an initial length x_0 and the displacement x, the extension of the string is given by $(x_0 - x)$.

Equating the two equations for F:

$$mg = k(x_0 - x)$$

Comparing this to the linear equation $y = mx + c$, we get:

$$y = mg$$

$$x_{\text{axis}} = (x_0 - x)$$

Therefore, the spring constant k corresponds to the gradient of the slope.

Imports and Functions

All calculations were done in Python. Starting with imports:

import matplotlib.pyplot as plt from uncertainties import ufloat import numpy as np from scipy import odr